Project Design Phase
Solution Architecture

Date	6th March 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Solution Architecture

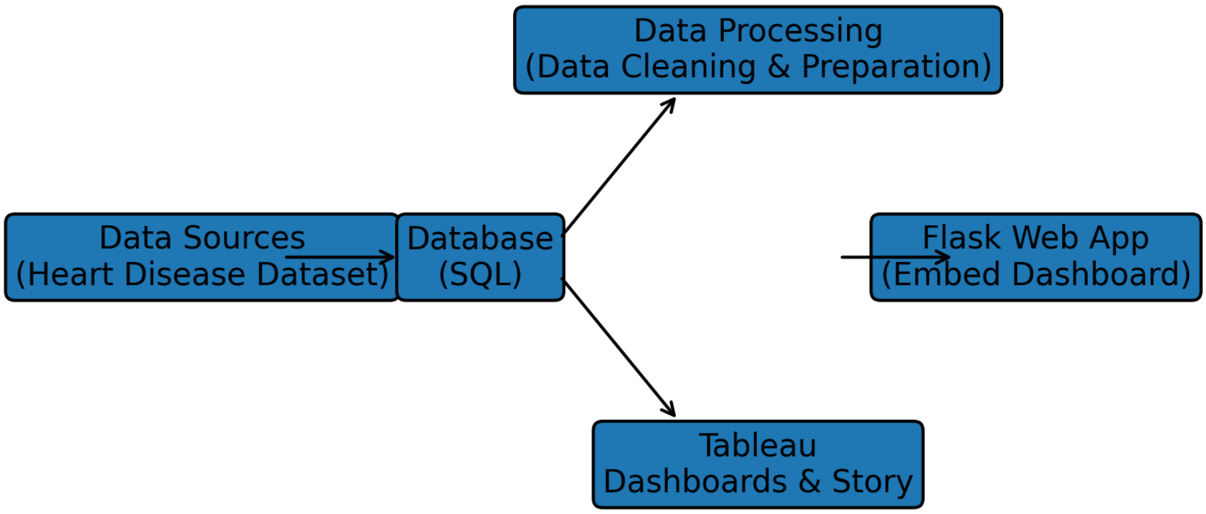
This project aims to build an interactive data analytics platform that analyzes heart disease data and presents meaningful insights through dashboards. The system integrates a database, data processing layer, Tableau visualization, and a web interface using Flask to deliver insights to healthcare professionals, policymakers, and individuals.
Efficient collection and storage of heart disease datasets.
Perform SQL operations for querying and managing patient health data.
Data preparation and transformation for accurate visualization.
Interactive dashboards built using Tableau to analyze risk factors such as age, BMI, cholesterol, and lifestyle habits.
Web integration using Flask to embed Tableau dashboards into a user-friendly interface.
Support decision-making for doctors, government health departments, and individuals.

System Components

Data Source: Heart disease dataset containing patient demographics, clinical indicators, and lifestyle factors.
Database Layer: Data stored and managed using SQL databases.
Data Processing Layer: Data cleaning, transformation, and preparation.
Visualization Layer: Tableau dashboards and stories for analysis.
Web Integration Layer: Flask-based web application embedding dashboards.
End Users: Doctors, policymakers, and individuals analyzing health risks.

Architecture Diagram

See the attached architecture diagram image.


Data Processing
(Data Cleaning & Preparation)
Data Sources
(Heart Disease Dataset)
Database
(SQL)
Flask Web App
(Embed Dashboard)
Tableau
Dashboards & Story